QWeatherRiskF — Merged District-Year Dataset: Scientific Data Guide (v1b)

Prepared 11 Oct 2026 · Dataset file: `merged_district_year_v1b.csv` *· Status: exploratory data guide, v1*

1. What this dataset is, in one paragraph

Each row describes **one Andhra Pradesh district in one year**, and holds two kinds of information side by side: **what the farmers produced** (area, production and yield for 23 crops, from the ICRISAT District Level Database) and **what the weather was** (rainfall from the IMD gridded product and temperature, humidity, wind and solar radiation from ERA5-Land, summarised into seasonal windows). The dataset exists to answer one question for our project: *when the weather of a season was unusual, did the district's crop yield move with it, and in which crops?*

Property	Value
Rows	65 (13 districts × 5 years: 2015–2019)
Columns	160
Join key	`district_code` + `year` (official AP district codes 502–523)
Districts	The 13 historical Andhra Pradesh districts
Crop source	ICRISAT District Level Database, *unapportioned* version (districts as they were reported)
Weather source	IMD 0.25° daily gridded rainfall; ERA5-Land daily reanalysis aggregated to districts
Missing data	Never filled. Missing values are blank (NaN).

2. How the dataset was built

1. **Weather file (daily).** For 13 districts and 2,557 days (1 Jan 2015 to 31 Dec 2021), each day has district-level rainfall (IMD) and temperature, humidity, wind and solar radiation (ERA5-Land). Srikakulam's rainfall is blank because only 75.1% of its area has valid IMD grid cells, and the earlier processing withheld districts below 90% coverage.

2. **Daily to yearly.** For every district and year, the daily series is summarised over six *windows* (for example the Kharif season) using 13 agronomically meaningful *features* (for example total rainfall, longest dry spell). That gives 78 weather columns plus one coverage column.
3. **Crop file (yearly).** ICRISAT reports area, production and yield per crop per district per year. Negative values are ICRISAT missing codes (−1) or an entry error (−500), and were set to blank.
4. **District codes.** Both files used different spellings (for example `Y.S.R.` and `Kadapa YSR`; `Spsr Nellore` and `S.P.S.Nellore`). Names were normalised and mapped to the official codes, and the mapping was checked to be one-to-one in both files. ICRISAT's own district code was dropped, because it clashes with the official one (ICRISAT 503 is Vizianagaram, but official 503 is Chittoor).
5. **Merge.** The two tables were joined one-to-one on `district_code` and `year` (65 of 65 rows matched).

3. Column groups at a glance

Group	Columns	Meaning
Identifiers	6	Which district and year
Crop: area, production, yield	69 (23 crops × 3)	What was grown and harvested
Crop: area only	6	Horticulture and fodder area
Weather: quality	1	How much of the district the rainfall data covers
Weather: windows × features	78 (6 × 13)	What the weather was, season by season
Total	**160**	

4. Identifier columns

Column	Meaning
`district_code`	Official AP district code (502 Anantapur, 503 Chittoor, 504 YSR, 505 East Godavari, 506 Guntur, 510 Krishna, 511 Kurnool, 515 SPR Nellore, 517 Prakasam, 519 Srikakulam, 520 Visakhapatnam, 521 Vizianagaram, 523 West Godavari). The key used for every join.
`district_name`	Official English name for that code
`year`	The ICRISAT year. Evidence so far suggests it means the **crop year starting in that year** (for example 2017 = 2017-18), but this must be confirmed in the ICRISAT documentation.
`State Code`, `State Name`	ICRISAT's state identifiers (Andhra Pradesh). Constant.
`crop_dist_name`	ICRISAT's spelling of the district, kept for traceability

5. Crop columns (75)

5.1 Where they come from

The ICRISAT District Level Database compiles district agricultural statistics that Indian state agriculture departments report (area sown, production from crop-cutting estimates and yield). Production and yield are therefore *estimates*, not weighings of every field. We use the **unapportioned** version: data are as reported for each district in the year, so Vizianagaram (formed from parts of Srikakulam and Visakhapatnam) and Prakasam appear as their own rows. The apportioned version instead folds new districts back into their parents, which would not match our weather districts.

5.2 The three-column pattern

Every crop has three columns:

Column pattern	Unit	Meaning
`<CROP> AREA (1000 ha)`	thousand hectares	Land sown to the crop that year. The **exposure** measure: how much of the district's farming depends on this crop.
`<CROP> PRODUCTION (1000 tons)`	thousand tonnes	Harvested output
`<CROP> YIELD (Kg per ha)`	kilograms per hectare	Production ÷ area × 1000. **The outcome weather is expected to influence.**

Yield is the scientifically important column. Area changes because of farmer decisions and market prices, while yield per hectare is closer to the crop's physiological response to the season. In the data, yield equals production ÷ area to within 2% for rows with production ⩾ 20 thousand tonnes (the rest is rounding).

5.3 The crops

Crop	Season character in AP	Note for our use
RICE	Kharif and Rabi combined	Largest crop (about 2.2 million ha). Mostly irrigated, so weak response to local rain.
WHEAT	Negligible in AP	Ignore
KHARIF SORGHUM, RABI SORGHUM	Kharif / Rabi	Kharif values are partly imputed. Rabi sorghum is small and concentrated in a few districts.
SORGHUM	Sum of the two	Composite
PEARL MILLET, FINGER MILLET	Mostly Kharif	Small
MAIZE	Kharif and Rabi	Both seasons mixed in one annual value
BARLEY	Not grown	Mostly missing codes
CHICKPEA	**Rabi**	Grown on residual soil moisture after the monsoon. The cleanest Rabi crop, but only in about 6 districts.

Crop	Season character in AP	Note for our use
PIGEONPEA	Kharif	Out of Rabi scope
MINOR PULSES	Kharif and Rabi	A bundle (blackgram, greengram, horse gram and others)
GROUNDNUT	Kharif (rainfed) and Rabi (irrigated)	Strongest rainfall response, because it is largely rainfed
SESAMUM, RAPESEED AND MUSTARD, SAFFLOWER, CASTOR, LINSEED, SUNFLOWER, SOYABEAN	Minor oilseeds	Tiny areas; many values imputed or missing
OILSEEDS	Total of all oilseeds	Includes oilseeds not listed separately
SUGARCANE	Long-duration	Yield jumps about 10× in 2018 (a unit or definition change, to be verified). Not comparable across years.
COTTON	Kharif	2019 yields look implausible (about 4–5× lower in every district); area and production are rounded to whole numbers
FRUITS, VEGETABLES, FRUITS AND VEGETABLES, POTATOES, ONION, FODDER	Horticulture, fodder	**Area only**: no production or yield columns

Season labels come from general agronomic knowledge and should be verified against ICRISAT documentation.

5.4 Data-quality rules applied or required

Rule	Reason
`-1` → blank (471 cells)	ICRISAT's "not reported" code, not a real value
`-500` → blank (1 cell)	A data-entry error (Nellore 2018 Kharif sorghum yield)

Rule	Reason
Ignore district-crops with area under 5,000 ha	Rounding and imputation make small-area yields unreliable (applied at analysis time)
Flag near-identical yields across 3 or more districts	Sunflower and chickpea in small districts share suspiciously equal values, so they are not independent measurements
Flag year-to-year breaks in a crop's median yield	Catches the sugarcane 2018 and cotton 2019 problems

6. Weather columns (79)

6.1 Where the numbers come from, scientifically

Rainfall: IMD gridded product. The India Meteorological Department builds a daily map of rainfall on a 0.25° × 0.25° grid (cells of about 25 km) by interpolating readings from its rain-gauge network. It is the most direct observation of rain available at this scale, but it is a *gridded estimate*, which smooths out small, intense storms and is defined only over land cells. A district's value is the area-weighted average of the grid cells that fall inside it. The IMD rainfall "day" is believed to run from 08:30 to 08:30 IST (to be verified).

Temperature, humidity, wind, solar radiation: ERA5-Land. ERA5-Land is a *reanalysis*: a weather and land-surface model re-run over past decades and constrained by observations, giving a complete, gap-free record at about 9 km resolution. The variables used are 2 m air temperature (daily maximum and minimum from hourly values), humidity (derived from temperature and dew point), 10 m wind speed and surface solar radiation. Reanalysis is not a measurement: values can differ from station readings, and daily extremes are often slightly muted. The file's own quality label reads `STRUCTURE_OK_PHYSICAL_QA_PENDING`: the structure is correct and our review is the physical check. Details still to be confirmed are whether days were cut at UTC or IST midnight and exactly how humidity was derived.

6.2 The coverage column

Column	Meaning
`rain_valid_area_fraction`	Share of the district's area covered by valid IMD cells (constant per district). Between 0.75 (Srikakulam, where a coastal strip has no IMD data) and 1.0. Tells us how representative the rainfall number is.

6.3 The six windows

A yearly average hides what matters to a crop, so each variable is summarised over **windows** that follow the agricultural calendar. Every window is a fixed date range tied to the ICRISAT year *Y*, and a window that is incomplete gives blank values rather than a guess.

Prefix	Dates	Agricultural meaning
`annual_cal__`	1 Jan – 31 Dec of Y	Calendar year
`annual_crop__`	1 Jun Y – 31 May Y+1	Crop year (monsoon through summer)
`kharif__`	1 Jun – 31 Oct of Y	South-west monsoon season; Kharif crops grow and mature
`ne__`	1 Oct – 31 Dec of Y	North-east monsoon, the main rain source for coastal AP and Rayalaseema in the late year. It supplies water for late Kharif crops and soil moisture for Rabi sowing.
`rabi_next__`	1 Nov Y – 31 Mar Y+1	Rabi season if year *Y* is the crop year starting in *Y*
`rabi_prev__`	1 Nov Y−1 – 31 Mar Y	Rabi season if year *Y* is the year of harvest. Blank for 2015, which would need November 2014.

The `kharif` and `ne` windows overlap in October by design. Each column name is `window__feature`, for example `kharif__rain_sum`.

6.4 The 13 features (each appears once per window, 6 × 13 = 78 columns)

Feature	How it is computed	What it measures, and why it matters	Caveat
`rain_sum` (mm)	Sum of daily district rainfall	Total water supplied by rain. Rainfed crops such as groundnut respond strongly to shortfalls.	Irrigated crops, such as most rice, are partly insulated
`rainy_days` (days)	Days with rain ≥ 2.5 mm (IMD's rainy-day definition)	Frequency of rain. The same total can come from a few violent storms (much runs off) or many gentle rains (more is absorbed).	Gridding and district averaging may inflate counts, so use for comparing years, not absolute values
`longest_dry` (days)	Longest run of consecutive days below 1 mm	The classic agricultural drought indicator. Crops live on stored soil water for one to three weeks, so a long dry spell at a sensitive stage cuts yield.	Threshold is a placeholder. For the Rabi window it is season-long (median about 70 days), so it is only meaningful inside a growth-stage window.
`max_5day_rain` (mm)	Largest rainfall total in any 5 consecutive days	Intensity of the wettest spell: flooding, waterlogging, crop lodging, disease and harvest losses	Smoothed by district averaging
`tmax_mean` (°C)	Mean of daily maximum temperature	Daytime heat load, which affects photosynthesis and development speed	Mostly a district fingerprint
`tmin_mean` (°C)	Mean of daily minimum temperature	Night warmth. Warm nights raise respiration and can reduce grain filling.	Mostly a district fingerprint

Feature	How it is computed	What it measures, and why it matters	Caveat
`hot_days` (days)	Days with Tmax ≥ 35 °C	Heat stress. Many crops lose seed or grain set when flowering coincides with temperatures above roughly this level (crop-specific, to be verified).	**Placeholder threshold**
`cold_days` (days)	Days with Tmin ≤ 15 °C	Cold stress, relevant to Rabi crops at sensitive stages	**Placeholder threshold.** Zero everywhere in the Kharif window, so useless there.
`dtr_mean` (°C)	Mean of (Tmax − Tmin)	Diurnal temperature range. Large means clear, dry skies; small means cloud and humidity. It affects grain filling and disease.	Also a cloudiness proxy
`rh_mean` (%)	Mean relative humidity	Fungal disease pressure (high) and evaporative demand (low)	Derivation method to be confirmed
`solar_mean` (MJ/m²/day)	Mean surface solar radiation	The energy for photosynthesis, and the main driver of water loss from crops	Low in the cloudy monsoon
`wind_mean` (m/s)	Mean 10 m wind speed	Evaporative demand. Strong winds also cause lodging.	A daily mean misses gusts, so storm damage is under-represented
`gdd_sum` (°C·days)	Sum over days of max(Tmean − 10, 0), with Tmean = (Tmax + Tmin) / 2	Growing degree-days: accumulated heat that sets how fast a crop develops. Warm seasons shorten each growth stage.	Base temperature 10 °C is a placeholder; real bases differ by crop

6.5 Which features carry real year-to-year signal?

A diagnostic on this dataset showed that for the temperature, humidity, wind and degree-day features, only about 2–10% of the variation is year-to-year within a district. The rest is simply the district's fixed climate. Rainfall features carry much more time-varying signal (about 28–57% for totals, and 56–81% for dry-spell length and wettest-5-day rain). So in any model, raw temperature means mostly identify the district, and we must use anomalies relative to normals.

7. How the columns relate to each other

`weather in a window → crop growth and stress at that time → yield per hectare → production = area × yield`.

The weather columns are the *drivers*, yield is the *response*, area is the *exposure*, and production is a derived result. Irrigation, soil, variety, pests, prices and policy also affect yield, and none of them are in this table.

8. A worked example: Anantapur (district 502), year 2018

Anantapur is a dry, mostly rainfed district. The row for 2018, set against the district's own average over 2015–2019:

Column	2018 value	District 5-year average	Difference
`kharif__rain_sum`	281.5 mm	453.3 mm	−38%
`kharif__rainy_days`	28 days	41.8 days	−33%
`kharif__longest_dry`	22 days	19.4 days	slightly longer
`ne__rain_sum`	50.7 mm	156.3 mm	−68%
`annual_crop__rain_sum`	359.8 mm	576.9 mm	−38%
`rabi_next__hot_days`	42 days	32.4 days	more heat in the following Rabi season
`GROUNDNUT AREA (1000 ha)`	490.6	482.8	about the same

Column	2018 value	District 5-year average	Difference
`GROUNDNUT YIELD (Kg per ha)`	**360**	658.4	**−45%**
`GROUNDNUT PRODUCTION (1000 tons)`	176.6	298.3	−41%
`CHICKPEA YIELD (Kg per ha)`	248	481.0	−48%
`RICE YIELD (Kg per ha)`	2,622	2,846	−8%

Reading it scientifically.

1. **The season was dry.** Kharif rain was 38% below this district's average, with fewer rainy days. The north-east monsoon, which follows, delivered about a third of its usual rain. Across all districts, mean Kharif rain in 2018 was 561 mm against a five-year average of about 716 mm (−22%), so 2018 was a poor year statewide.
2. **The rainfed crops fell with the rain.** Groundnut yield fell 45% while the sown area barely changed. Since farmers had already sown the crop, the shortfall is a *yield* effect, not a decision effect. Chickpea, which depends on moisture left in the soil after the monsoon, fell by a similar amount.
3. **The irrigated crop barely moved.** Rice fell only 8%, consistent with canal and borewell water cushioning it against low rain.
4. **The arithmetic holds.** Area × yield: 490.6 × 360 ≈ 176.6 thousand tonnes, which is the production column.

What this row cannot tell us. It shows a *seasonal association*, not which growth stage was hurt: a dry spell at flowering and one at harvest would look the same in a yearly yield. With only five years, and 2016 and 2018 both being poor years across the state, a large share of the pattern is shared by all districts, so this supports a hypothesis and its sign but cannot give a reliable effect size. Averages over five years are also a weak "normal".

9. Known limitations of this dataset

1. **Only 5 years (2015–2019)** for 13 districts. The effective number of independent years is about 4–5.
2. **Annual crop values mix Kharif and Rabi** for rice, groundnut, maize and pulses. No season-wise split is available.
3. **No crop-stage data.** Stage-specific effects must come from agronomic knowledge and cannot be learned here.
4. **Srikakulam rainfall is missing** because of a coastal land-mask gap in the IMD grid (ADR-013 records the provisional policy). Its rain features are blank.

5. **District averages smooth extremes**, so storms are muted.
6. **ERA5-Land is a reanalysis**, not station data, and its temperature has not yet been checked against IMD temperature.
7. **Several thresholds are placeholders** (rainy day, dry day, hot day, cold day, base temperature) until crop-specific evidence replaces them.
8. **Year convention is tentative**, and some crops have unit or reporting breaks (sugarcane 2018, cotton 2019).
9. **Yield depends on much more than weather:** irrigation, soil, variety, pests, prices and policy are absent.

10. Suggested next steps

1. Confirm the ICRISAT year convention from its documentation.
2. Extend the weather record back to at least 1990, so that proper long-term normals and anomalies can be computed.
3. Choose features per crop by agronomic hypothesis, not by correlation, and convert them to anomalies.
4. Report every result with its effective sample size.

Glossary

Term	Meaning
Window	A fixed date range over which daily weather is summarised
Anomaly	A value minus the long-term normal for that place
Reanalysis	A weather model re-run on past data and fitted to observations
Gridded product	Station data interpolated onto a regular map of cells
Kharif / Rabi	Monsoon season crops / winter-season crops
Growing degree-days	Accumulated heat above a base temperature, used to time crop development
Dry spell	Consecutive days with almost no rain
Unapportioned	District data as originally reported, without merging new districts into their parents